

UC-I — Vessel Traffic Management & Optimization

Workstream 2: AI / ML Tools (Tabular Description)

Tender Ref: GEM / 2026 / B / 7297343 · Submission: PoC Pilot, 07 Aug 2026 Columns per Bidder Briefing WS2: Use Case Solved · Training Data (Features) · Objective Function · Model Used · Rationale · Link to Model Weights · Validation Data · Accuracy

Disclosure (methodology honesty). UC-I engines are deliberately **deterministic, physics-based and explainable** for the PoC: under-keel clearance and tidal windows are safety-of-navigation computations where auditable formulas are preferred over black-box ML, and the shared corpus does not contain enough labelled vessel-call history to train credible arrival-time models. Where a component is rule-based, the "Model Weights" column points to its versioned configuration (coefficients) instead of learned weights; the production path to learned models is stated in each row.

#	Use Case Solved	Training Data (Features)	Objective Function	Model Used	Rationale	Link to Model Weights	Validation Data	Accuracy
1	DUKC / RTUKC — is there enough water under the keel for a transit, per channel reach	No training (physics). Inputs: static draft, charted depth per reach, tide height, speed (kt), block coefficient (container ≈ 0.65, bulk ≈ 0.80)	Gross UKC = charted depth + tide - (draft + squat); go if net UKC ≥ 1.0 m safety margin (marginal band 0.6 m)	Deterministic UKC formula with Barrass-type squat (squat ≈ Cb·V²/100, clamped 0–2.5 m); mixed semi-diurnal tide model (mean 2.6 m, amplitude 1.7 m)	Safety-critical computation must be auditable and explainable to the Deputy Conservator; parametric per vessel class; production upgrade = vendor DUKC service calibrated on survey + tide-gauge feeds	<code>src/dukc/ukc.ts</code> (versioned constants: UKC_SAFETY_MARGIN_M = 1.0, UKC_MARGINAL_BAND_M = 0.6)	Sensitivity analysis at draft ± 0.2 m / tide ± 0.1 m corner cases (<code>ukcSensitivity()</code>); replay against shared tide readings (188 obs)	Deterministic (exact inputs, confident expressions) / no-go light w/ sensitivity corner
2	Tidal-window computation & extension — feasible transit windows per call over 5 days	No training. Tide curve walked at 0.25 h steps over 120 h; controlling depth = min charted depth of inner/turning reaches ± siltation/dredging delta	Maximise usable window duration subject to net-UKC ≥ margin for the call's draft	Window-scan algorithm over the DUKC physics model (<code>tidalWindows()</code>)	Directly answers the briefing item "extend tidal window": the effect of dredging (+restore m) or siltation (-m) on window width is computed live	<code>src/dukc/ukc.ts</code> + channel model <code>src/map/portGeometry.ts</code> (reaches CH-OUTER 17.5 m ... CH-INNER 15.0 m controlling)	Cross-checked against published channel design depth from bathymetry title blocks (shared survey PDFs)	Deterministic window constraints under tests (scenarios)
3	Vessel TAT prediction (ETB→ETD horizon)	Feature vector from live state + levers: parcel TEU, draft vs terminal max, weather severity, rain mm/hr, net channel-depth delta, pilots/tugs down, extra arrivals, incident severity, berth-window extension	Minimise expected TAT error; output P50 with P10/P90 band (σ = 2 + 0.3·stressor h, ±1.28σ)	Transparent additive linear feature model (documented coefficients, e.g. base 34 h, +1 h/250 TEU, +1.5 h per m over 13 m draft ref, +5 h weather, +4 h per m depth loss)	Explainable per-factor contribution chart for operators; corpus lacks per-call ground-truth history for supervised training; production = gradient-boosted regressor retrained on accumulated actuals	<code>src/planning/tatPredict.ts</code> (TAT_MODEL coefficient block, versioned)	Calibration figures from JNPA public reference: TAT ≈ 1.83 d, berth stay ≈ 0.97 d, ~10–12 calls/day (assumptions register)	Report actual prediction KPI: forecast accuracy – MAF 100 trials live on dashboard (see V)
4	ETA/ETD → berth utilisation — occupancy, waiting distribution, ETA confidence	Berthing-plan intervals (planned vs actual), AIS positions & staleness	Berth occupancy % = occupied berth-hours / (berths × window); ETA band widens with horizon (0.06 h/h) + AIS staleness (0.05 h/min)	Deterministic analytics engine (<code>occupancyCalendar</code> , <code>waitingTimeDistribution</code> p50/p90, <code>etaDistribution</code>)	The utilisation question is measurement, not prediction; distributions (median/P90) rather than bare means per spec	<code>src/kpi/analytics.ts</code> , <code>src/kpi/formulas.ts</code>	Replay over ingested berthing-report data (corpus berthing PDFs → plan records)	Deterministic aggregated p50/p90 reports each f
5	Berth-plan optimisation / re-sequencing under disruption	Berth requests (LOA, draft, requested window, duration), berth compatibility, go-windows from tidal model	Minimise cost = 1·Σwait-h + 2·tide-misses + 0.5·berth-shifts (weights configurable)	Greedy heuristic optimiser — earliest feasible compatible berth, preferring requested berth (explicitly not globally optimal)	Transparent, instant re-plans during what-if demos; production = MILP/CP-SAT solver on the same request/constraint schema	<code>src/planning/optimiser.ts</code> (DEFAULT_WEIGHTS versioned)	Scenario suite M1–M10 regression (plan feasibility + cost deltas under each disruption)	Feasible guarantee constraints cost break (wait/t show)
6	JIT arrival / RTA advice with bunker & CO₂ delta	Distance to go, berth-ready time, go-window start, service speed 16 kn, fuel 3.2 t/h at service speed	RTA = max(berth ready, go-window start); speed = distance/available time; fuel = speed²; CO₂ = 3.114 t/fuel-t (IMO factor)	Deterministic physics/heuristic (labelled SIMULATED for commercial figures, USD 600/t bunker assumption)	Demonstrates the JIT KPI loop: slow-steam advice instead of anchorage wait, with quantified fuel/CO₂ saving	<code>src/planning/jit.ts</code> (constants versioned)	Cross-checked against IMO CO₂ factor and typical container-vessel consumption curves	Deterministic saving always preserved simulation deltas named assum
7	Port-craft assignment & conflict detection (pilot/tug/mooring)	Roster (9 craft: 3 pilots, 4 tugs, 2 mooring), status, response minutes, committed demand from plan	Detect near-term demand > serviceable supply; propose single-unit swap minimising response gap	Deterministic utilisation + conflict heuristic with recommendation engine (<code>detectConflicts</code> , <code>recommend</code>)	Craft shortage is a briefing edge case (EC-4); recommendations must be explainable to marine control	<code>src/components/reports/PortCraftBoard.tsx</code> , <code>src/sim/derive.ts</code>	Scenario M4 (2 pilots down) regression; roster from shared pilot-card & port-craft documents	Deterministic gap-closure minutes quoted simulation delta v nothing
8	Reactive confidence chain — tidal + port craft + vessels at sea → DUKC → confidence (briefing reactive item)	Live tide, roster state, AIS fleet state, causal DAG (23 nodes / 30 edges: weather→pilotage→queue→delay→TAT; tide→DUKC→deep-draft window)	Propagate lever/sensor changes through the causal graph; recompute downstream KPIs and DUKC confidence	Causal-graph propagation over the deterministic engines (<code>src/whatif/causalGraph.ts</code>) + workflow rules (e.g. wind ≥ 30 kn → hold pilotage + notify)	This is the "reactive twin" evidence: one upstream change visibly cascades to DUKC state, plan, craft roster and KPIs with an audit trail	<code>src/whatif/causalGraph.ts</code> , <code>src/workflow/rules.ts</code>	Guided-tour scenario suite (M1–M10) exercises every edge of the DAG	Quality: cascading correction demo live; e propagation step k

Production upgrade path (stated per marking criterion 2)

PoC component	Production replacement	Trigger
Additive TAT model	Gradient-boosted regressor (LightGBM) retrained monthly on accumulated ATA/ATB/ATD actuals	≥ 6 months of ingested call history
Greedy berth optimiser	MILP / CP-SAT berth-allocation solver, same schema	Post-award phase 1
Analytic tide curve	INCOIS/tide-gauge assimilated tide surface + vendor DUKC calibration	Live INCOIS agreement
ETA fixtures	Learned ETA model on AIS track history (port-approach segments)	Live AIS retention ≥ 3 months